

Self-powered 4-port USB 2.0 High-Speed hub with controlled impedance PCB design.

1 Features

- USB 2.0 High-Speed (480 Mbps)
- Self-powered architecture
- 4 downstream USB ports
- Controlled differential impedance routing
- 4-layer PCB with solid ground plane
- Local power decoupling network
- High-speed signal integrity optimized layout

2 Applications

- Embedded systems
- Industrial controllers
- Development platforms
- USB expansion modules
- Laboratory test equipment

3 Description

The USB 2.0 High-Speed 4-Port Hub is a self-powered USB 2.0 hub module designed to provide reliable High-Speed connectivity for multiple downstream USB devices. The hub expands a single upstream USB connection into multiple ports while maintaining full compliance with the USB 2.0 High-Speed specification, supporting data transfer rates up to 480 Mbps.

The design prioritizes signal integrity, stable power distribution, and robust PCB layout practices to ensure reliable operation under typical embedded and development environments. To support high-speed USB signaling, the board is implemented using a four-layer PCB architecture with a continuous ground reference plane and controlled impedance routing for all USB differential pairs.

The USB data lines are routed as matched differential pairs with a target impedance of **90 Ω $\pm 10\%$** , ensuring proper signal propagation and minimizing reflections, jitter, and electromagnetic interference. The routing strategy also includes controlled trace length matching and minimized discontinuities to preserve signal quality across the high-speed paths.

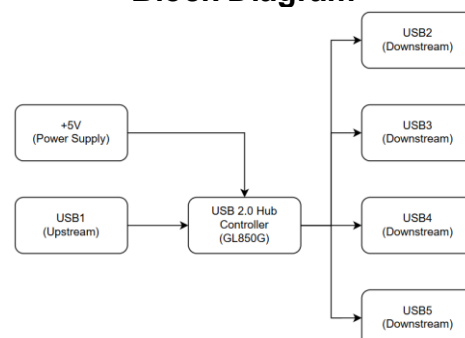
Power distribution is implemented using a dedicated power plane combined with local decoupling capacitors placed close to critical devices to reduce supply noise and maintain stable operating conditions. Ferrite bead isolation and bulk capacitance are used to improve filtering and reduce the propagation of switching noise across the power network.

The hub is configured for self-powered operation, allowing it to supply downstream USB ports from an external 5-V power source while maintaining proper upstream communication with the host system. This architecture improves power stability and ensures reliable device enumeration even when multiple peripherals are connected.

The board layout also incorporates stitching vias between ground planes to provide low-impedance return current paths and reduce loop area for high-speed signals. These techniques contribute to improved electromagnetic compatibility (EMC) and more predictable high-speed signal behavior.

The USB 2.0 High-Speed 4-Port Hub is intended for use in embedded systems, development platforms, laboratory setups, and custom electronic equipment where additional USB connectivity is required. The design emphasizes practical implementation of high-speed digital design principles and serves both as a functional expansion module and as a reference example of controlled-impedance USB routing on a multilayer PCB.

Block Diagram



4 Device Information

Parameter	Value
Product Type	USB 2.0 Hub Module
Ports	4 Downstream
Data Rate	480 Mbps
Power Mode	Self-Powered
PCB Layers	4
Dimensions	75mm X 50mm

5 PCB Stackup

Layer	Function
L1	High-speed signals
L2	Ground plane
L3	Power plane
L4	Control / low speed

6 Layout Guidelines

- Maintain 90Ω differential impedance for USB data pairs
- Match differential pair lengths within 0.5 mm
- Route differential pairs over a continuous ground plane
- Avoid stubs and sharp angles
- Place stitching vias near signal transitions between layers

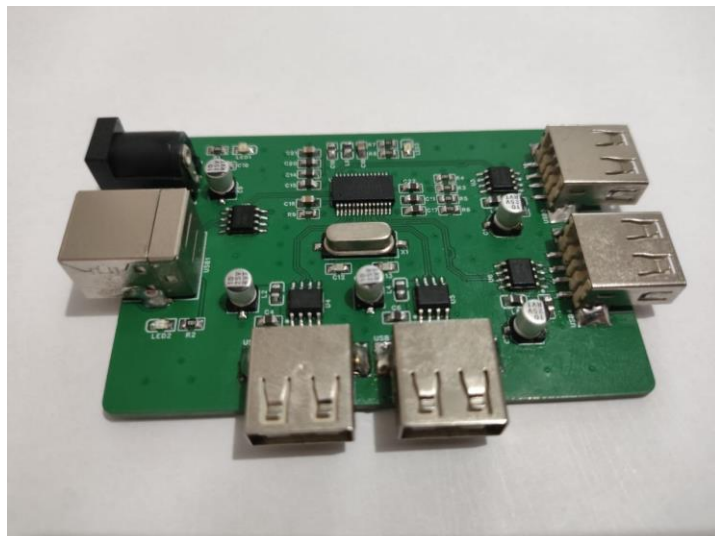


Figure 1. PCB Assembled

7 Schematic

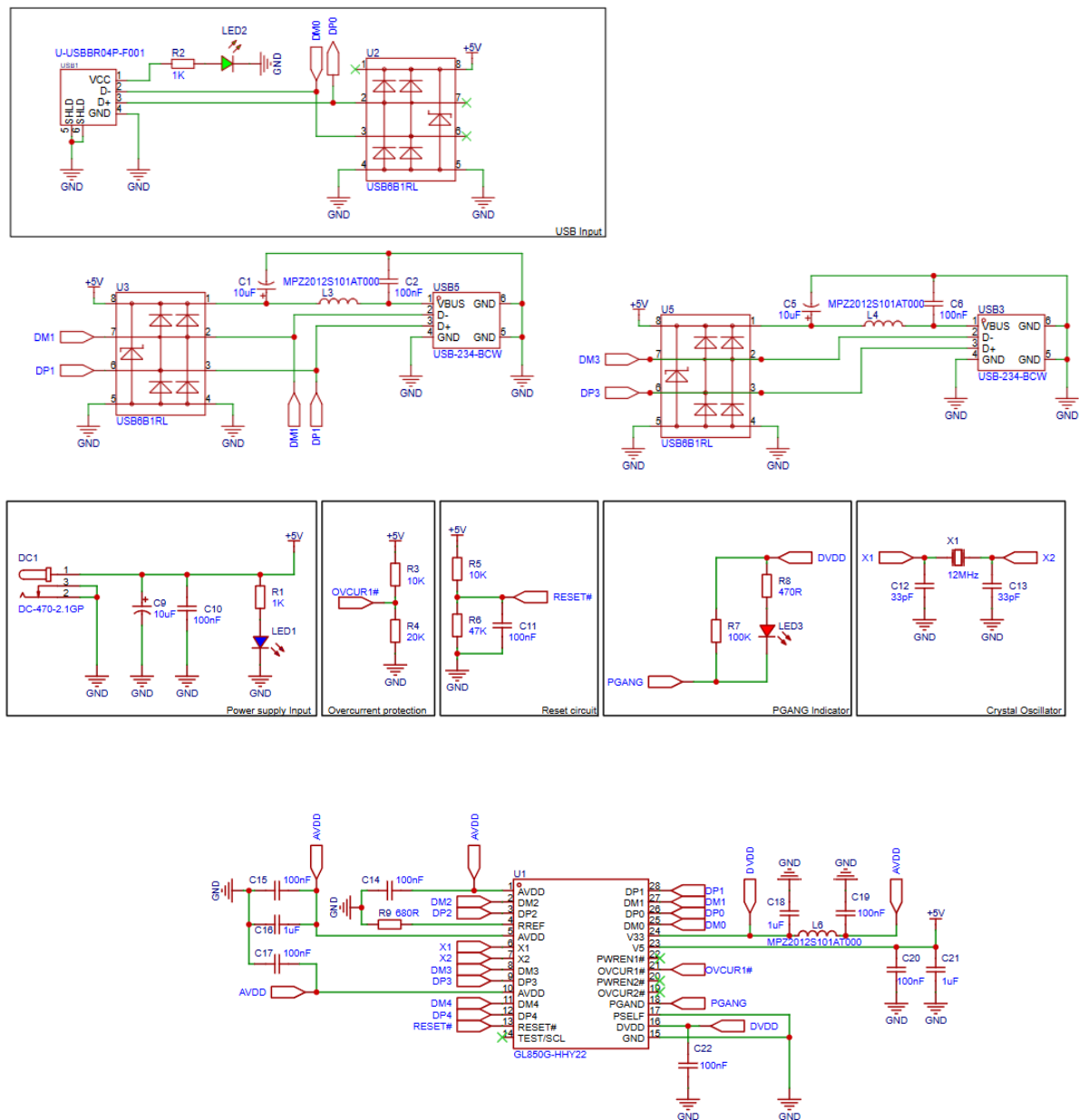


Figure 2. Schematic

8 BOM (Bill of Materials)

Name	Designator	Quantity
DC-470-2.1GP	DC1	1
GL850G-HHY22	U1	1
U-USBBR04P-F001	USB1	1
Crystal 12MHz	X1	1
Capacitor 1uF 50V 0805 X7R	C16,C18,C21	3
LED 0805	LED1,LED2,LED3	3
USB-234-BCW	USB2,USB3,USB4,USB5	4
Capacitor 10uF 50V 0805 X7R	C1,C3,C5,C7,C9	5
Ferrite Bead 100Ω@100MHz 0805 4A 20mΩ	L2,L3,L4,L5,L6	5
USB6B1RL	U2,U3,U4,U5,U6	5
Resistor 1K 0805	R1, R2	2
Resistor 10K 0805	R3, R5	2
Resistor 20K 0805	R4	1
Resistor 47K 0805	R6	1
Resistor 100K 0805	R7	1
Resistor 470R 0805	R8	1
Resistor 680R 0805	R9	1
Capacitor 33pF 50V 0805 X7R	C12,C13	2
Capacitor 100nF 50V 0805 X7R	C2,C4,C6,C8,C10,C11,C14,C15,C17,C19,C20,C22	12

9 PCB Layout

Top Layer

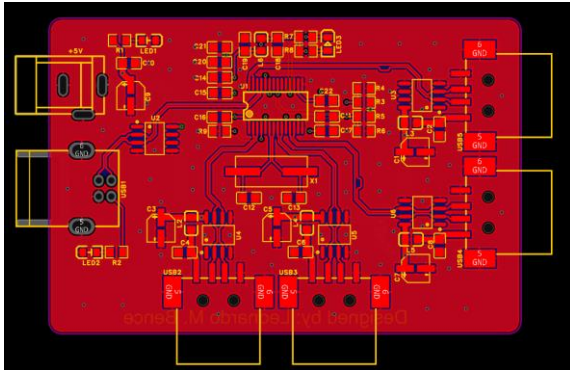


Figure 3. Top Layer

Inner 1

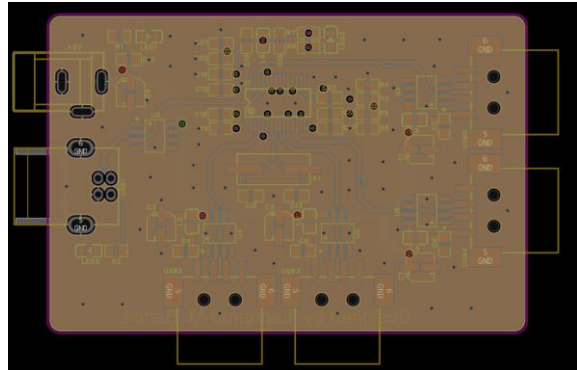


Figure 4. Inner 1

Bottom Layer

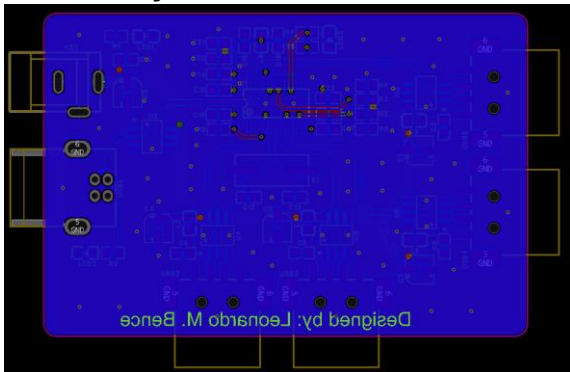


Figure 5. Bottom Layer

Inner 2

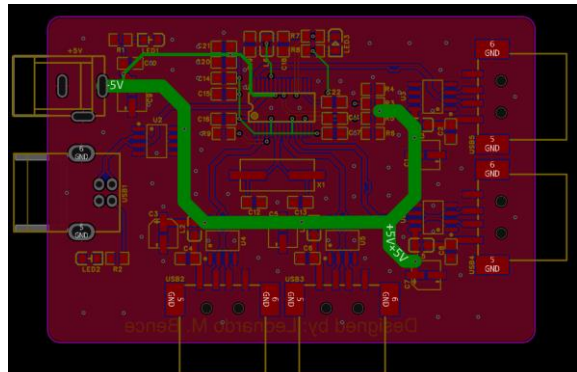


Figure 6. Inner 2

10 Mechanical Information

Parameter	Value
Length	75 mm
Width	50 mm
Thickness	1.6 mm
Material	FR4

11 Electrical Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Input Supply Voltage	External power supply	4.75	5.0	5.25	V
Input Supply Current	No downstream devices	—	60	120	mA
Maximum Downstream Current (per port)	USB 2.0 specification	—	—	500	mA
Total Output Current	All ports combined	—	—	2	A
USB Data Rate	High-Speed mode	—	480	—	Mbps
Differential Impedance (USB D+/D-)	Controlled impedance routing	81	90	99	Ω
Operating Temperature	Ambient	0	25	70	°C
Storage Temperature		-40	—	85	°C
ESD Protection (USB ports)	Contact discharge	—	—	± 8	kV
ESD Protection (USB ports)	Air discharge	—	—	± 15	kV

12 3D Image of the PCB

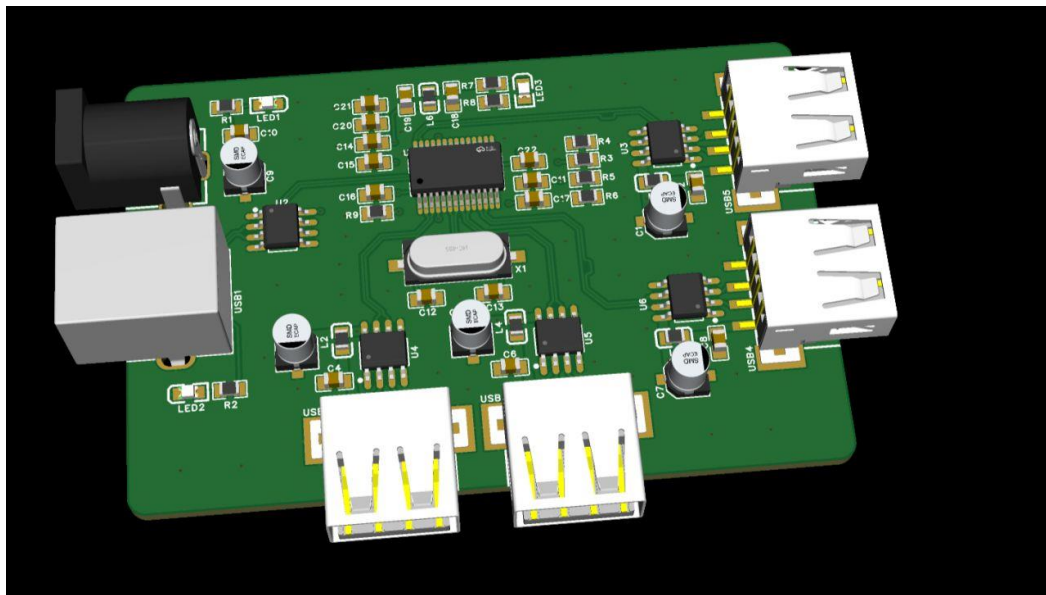


Figure 7. 3D PCB Image

13 USB Flash Drive File Transfer Test

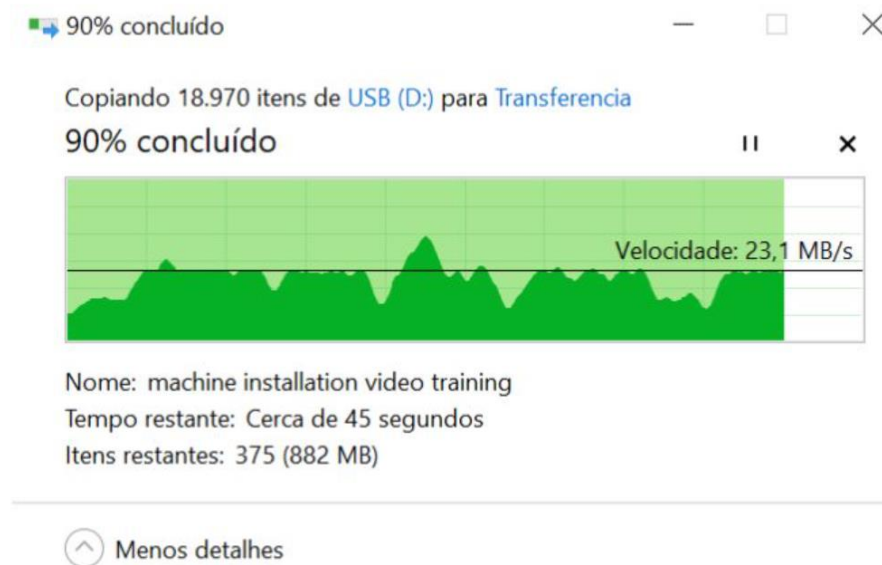


Figure 8. USB Flash Drive File Transfer Test

Note: Example of a file transfer test performed through the USB hub using a flash drive connected to a downstream port. The measured transfer rate (~23 MB/s) is consistent with typical USB 2.0 High-Speed storage device performance.

Disclaimer

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